Construction Chemicals

Product Datasheet
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Protective Coating

O-Guard EP

Solvent-Free Polyamine Epoxy Lining for Potable Water Tanks



Product Description

O-Guard EP is a solvent-free, polyamine-cured epoxy coating for lining potable water tanks and other concrete and steel structures. It is suitable for contact with drinking water and food, resists chemicals and corrosive vapours, and gives a glossy, easy-to-clean surface. An anti-slip finish is optional.

Uses

- Potable water tanks and reservoirs.
- Chemical and wastewater tanks.
- Industrial concrete and steel tanks.
- Car parks and bridge decks.
- Pharmaceutical and textile plants, basements and power stations.

Advantages

- Suitable for potable water contact.
- Solvent-free, 100 % solids.
- Resists chemicals and corrosive vapours.
- Glossy, easy to clean.
- Can be used in lamination systems.

Application

1. **Surface Preparation**
   Concrete ≥ 25 N/mm², pull-off ≥ 1.5 N/mm², ≥ 28 days, moisture < 4 %; prime with O-Prime ST.
2. **Mixing**
   Add Part B to Part A (200 : 30 by weight) and mix 2 – 3 min with a slow-speed mixer.
3. **Application**
   Apply by roller or airless spray at 0.25 – 0.30 kg/m² per coat; second coat within 48 hours.

Consumption

0.25 – 0.30 kg/m²
per coat (approx. 200 µm)
Two coats for tank linings.

Packaging

5 kg sets: Part A 4.35 kg in pail + Part B 0.65 kg in jerrycan.

Storage & Shelf Life

12 months in original, unopened packaging. Store dry at +5 °C to +35 °C, away from direct sunlight.

Technical Data

Typical values at 25 °C

Property	Typical Value	Test Method
Mixing ratio (A : B)	200 : 30 by weight — pre-weighed sets, mix as recommended	—
Specific gravity	1.57	—
Solids content	100 %	—
Flash point	145 °C	—
Pot life	40 min	—
Recoat	24 hours (max. 48 hours)	—
Full cure	7 days	—
Heat resistance	up to 80 °C	—
Colours	On request	—

Consumption

Layer	Product	Consumption
Primer	O-Prime ST	0.18 – 0.25 kg/m²
Body coat	O-Guard EP	0.25 – 0.30 kg/m²
Top coat	O-Guard EP	0.25 – 0.30 kg/m²

Consumption is theoretical and varies with substrate porosity, profile and application method.

Application Conditions

- Ambient and substrate temperature: +10 °C to +35 °C.
- Substrate at least 3 °C above dew point.
- Relative humidity: max. 85 %.
- Concrete ≥ 25 N/mm², pull-off ≥ 1.5 N/mm², ≥ 28 days, moisture < 4 %.

Application Notes

- Mix complete sets.
- Apply the second coat within 48 hours.
- Allow 7 days full cure and wash the tank with clean water before filling with potable water.
- Broadcast fine quartz sand on floors where anti-slip is required.

Health & Safety

- Wear chemical-resistant gloves, safety goggles and overalls.
- Uncured resin may cause skin sensitisation; avoid skin and eye contact.
- Ensure good ventilation in enclosed areas.

First Aid & Cleaning

- Eyes: rinse immediately with plenty of clean water for at least 15 minutes and seek medical advice.
- Skin: remove contaminated clothing and wash with soap and water; do not use solvents on skin.
- Inhalation: move to fresh air; seek medical advice if symptoms persist.
- Ingestion: do not induce vomiting; seek medical advice immediately.
- Clean tools with the recommended thinner before the material cures.